AmazonKart Black Friday Infrastructure

Secure, Reliable and Cost-Conscious AWS Infrastructure

Final Architecture, RCA, Disaster Recovery and Project Documentation

Project	AmazonKart - Black Friday Infrastructure Challenge
AWS Region	US East (N. Virginia) - us-east-1
Production OS	Amazon Linux 2023
DR RTO Test	3 minutes 9 seconds - PASS (<15 min target)
Document date	30 August 2026

1. Executive Summary

AmazonKart required a secure, reliable and cost-conscious production environment for a high-traffic Black Friday event. The implementation was constrained to the challenge's permitted AWS services: Amazon EC2, Amazon VPC, Amazon S3, AWS IAM and Amazon CloudWatch. The solution uses a custom VPC, hardened public web tier, isolated private subnet, private S3 storage, least-privilege IAM, centralized monitoring/logging, automated midnight backups and a tested AMI + S3 disaster-recovery procedure.

The recovery path was functionally tested by launching a new EC2 instance from the Golden AMI, verifying Apache and the website, validating public browser access, and restoring website data from S3. A separate timed recovery test completed in 3 minutes 9 seconds, meeting the stated RTO target of less than 15 minutes.

2. Architecture Diagram

```
                              INTERNET USERS
                                   |
                             HTTP 80 / HTTPS 443
                                   |
                          +-------------------+
                          | Internet Gateway  |
                          +---------+---------+
                                    |
                     AmazonKart-VPC 10.0.0.0/16
        +---------------------------+---------------------------+
        |                                                       |
+-------v----------------------+                 +--------------v-------------+
| Public Subnet 10.0.1.0/24   |                | Private Subnet 10.0.2.0/24 |
| AmazonKart-Public-Subnet-1   |                 | AmazonKart-Private-Subnet-1|
| Public NACL                  |                 | Private NACL                |
+--------------+---------------+                 +----------------------------+
               |
      +--------v-----------------+
      | EC2 Production Web Server |
      | AmazonKart-Prod-Web01     |
      | Amazon Linux 2023 / httpd |
      +-----+---------------+-----+
            |               |
            | IAM Role      | CloudWatch Agent
            |               |
     +------v------+    +---v------------------------------+
     | Amazon S3   |    | Amazon CloudWatch                |
     |             |    | CPU / Memory / Disk              |
     | Static      |    | Alarms / Dashboard               |
     | private     |    | Apache/Auth/System centralized logs|
     +-------------+    +----------------------------------+
     | Backup      |
     | daily/      |
     +------+------+
            |
       12:00 AM IST
       automated backup

DISASTER RECOVERY:
Production EC2 -> Golden AMI -> New DR EC2 -> S3 latest backup restore -> HTTP 200
Timed RTO test: 3m 09s (<15 min) - PASS
```

3. Network Design and Security

Component	Name / ID	CIDR / Scope	Purpose

VPC	AmazonKart-VPC	10.0.0.0/16	Custom production VPC
Public subnet	AmazonKart-Public-Subnet-1	10.0.1.0/24	Production EC2 / internet-facing tier
Private subnet	AmazonKart-Private-Subnet-1	10.0.2.0/24	Isolated private tier
Internet Gateway	igw-05de8d6969ecff270	-	Attached to AmazonKart-VPC
Public NACL	acl-0e4b7edca14577380	Public subnet	SSH restricted; HTTP/HTTPS; ephemeral return traffic
Private NACL	acl-042bd5c057b67bf13	Private subnet	Separated from public NACL; restrictive subnet-level control
Web Security Group	sg-0afcf9ae96a2fd000	EC2	Inbound 22 restricted to /32; 80/443 public

3.1 Final Security Group Inbound Rules

- SSH TCP/22 - restricted to the trusted administrator public IP /32.
- HTTP TCP/80 - 0.0.0.0/0 for the public website.
- HTTPS TCP/443 - 0.0.0.0/0 as an allowed web port.
- No unrestricted SSH rule remains.

4. EC2 Production Web Tier

Instance: AmazonKart-Prod-Web01

Instance ID: i-01d773eccc68d0c5f

Instance type: t3.micro

AMI: Amazon Linux 2023 (source AMI ami-0332d564d76dbd8d6)

Private IP: 10.0.1.209

Web server: Apache httpd

Website path: /var/www/html

SSH: Public-key authentication enabled; password authentication disabled

IMDS: IMDSv2 required

5. IAM and Least Privilege

Four functional IAM teams were created instead of granting AdministratorAccess broadly. Console users were placed in their corresponding groups and MFA was configured. The EC2 production role provides only the S3 access required by the application and backup/recovery workflow.

IAM Group	Access Intent
AmazonKart-Operations	EC2 operations, CloudWatch read, backup access
AmazonKart-Security	IAM / VPC / EC2 / S3 / CloudWatch security visibility
AmazonKart-Audit	Read-only audit visibility
AmazonKart-Intern	Limited EC2 and CloudWatch read access

6. Amazon S3

6.1 Static Content Bucket

- Bucket: amazonkart-static-2026-28-08
- Bucket Versioning: Enabled
- Default encryption: SSE-S3
- Block all public access: On
- S3 static website hosting: Disabled
- EC2 retrieves private static content using its IAM role. profile.jpeg was copied to /var/www/html/static/profile.jpeg and served by Apache with HTTP 200.

6.2 Backup Bucket

- Bucket: amazonkart-backup-2026-28-08
- Backup prefix: daily/
- Production role permissions include ListBucket and PutObject; GetObject is included for disaster-recovery restore.
- Website source backed up: /var/www/html

7. Automated Daily Backup

The EC2 host timezone was set to Asia/Kolkata so that the cron expression maps directly to midnight IST.

```
0 0 * * * /usr/local/bin/amazonkart-backup.sh
```

The backup script creates a timestamped tar.gz archive of /var/www/html, uploads it to the S3 backup bucket under daily/, records success/failure in /var/log/amazonkart-backup.log, and removes the temporary local archive after successful upload.

- Manual verification showed a successful backup object: amazonkart-web-2026-08-30_11-55-56.tar.gz.

8. CloudWatch Monitoring, Alarms and Logging

Resource	Metric	Threshold	Period	Alarm
CPU	EC2 CPUUtilization	>75%	5 minutes	AmazonKart-CPU-High-75
Memory	mem_used_percent	>80%	1 minute	AmazonKart-Memory-High-80
Root disk	disk_used_percent	>85%	1 minute	AmazonKart-Disk-High-85

Dashboard: AmazonKart-Production-Dashboard with separate CPU, memory and root disk widgets.

8.1 Centralized Log Groups

CloudWatch Log Group	Source
/amazonkart/production/apache/access	/var/log/httpd/access_log
/amazonkart/production/apache/error	/var/log/httpd/error_log
/amazonkart/production/auth	/var/log/secure (Amazon Linux 2023 authentication equivalent)
/amazonkart/production/system	/var/log/messages

Retention: 1 month (30 days) for all four log groups.

Note on the challenge: the requirement text requested SNS email notification, while the strict allowed-service list excluded Amazon SNS. The implementation therefore keeps CloudWatch alarms configured without SNS actions and records the requirement conflict rather than using a prohibited service.

9. Disaster Recovery Plan and Evidence

9.1 Recovery Strategy

The DR strategy combines an EC2 Golden AMI for rapid server reconstruction with S3 backups for website-data recovery. This avoids AWS Backup, Auto Scaling, load balancers and other services excluded by the challenge.

1. Launch a replacement EC2 instance from AmazonKart-Prod-Golden-AMI (ami-05f02a718c6d21b63).
2. Use t3.micro, AmazonKart-Public-Subnet-1, public IPv4, AmazonKart-Web-SG, AmazonKart-Prod-Key and AmazonKart-EC2-Production-Role.
3. Wait for the instance to become healthy and verify SSH access.
4. Verify httpd is active and curl http://localhost returns HTTP 200.
5. Verify /static/profile.jpeg returns HTTP 200 and test the public browser endpoint.
6. List the S3 backup bucket, download the latest daily/*.tar.gz backup, preserve the current web directory, and extract the backup under /var/www.

7. Re-test the website and static content. Confirm HTTP 200 and browser rendering.

9.2 DR Test Results

Test	Evidence	Result
Golden AMI	ami-05f02a718c6d21b63	PASS
Replacement EC2 SSH	Successful public-key login	PASS
Apache service	active	PASS
Local website test	HTTP/1.1 200 OK	PASS
Static image test	HTTP/1.1 200 OK	PASS
Public browser test	Website and image rendered	PASS
S3 backup listing	Successful	PASS
S3 GetObject restore	Successful after least-privilege policy included GetObject	PASS
Post-restore website	HTTP/1.1 200 OK	PASS
Timed RTO test	3 minutes 9 seconds (<15 min target)	PASS

10. RCA - Disaster Recovery Test

Incident / Scenario	Simulated loss of the production EC2 web server requiring rapid service restoration.
Business impact	Potential loss of the AmazonKart web service during the Black Friday traffic window.
Root cause category	Infrastructure instance failure simulation; no production outage was intentionally induced.
Detection	EC2/CloudWatch health and service checks; website availability verified by HTTP and browser tests.
Recovery action	Launch replacement from Golden AMI, attach existing network/security/IAM configuration, verify Apache, restore latest website backup from private S3, and validate HTTP 200.
Issue discovered during test	The EC2 role could list the backup bucket but initially could not download the backup object (403

	HeadObject).
Root cause of restore issue	s3:GetObject was not originally available for the backup daily/* object path.
Corrective action	Added s3:GetObject to the existing least-privilege backup object statement and repeated the restore successfully.
RTO result	3 minutes 9 seconds in the timed AMI recovery test - PASS against <15 minute target.
Preventive actions	Maintain Golden AMI after material configuration changes; retain daily S3 backups; periodically test restore; monitor alarms/logs; keep IAM permissions minimal but recovery-capable.

11. Security Controls Summary

- Only required inbound ports 22, 80 and 443 are configured on the web Security Group.
- SSH is restricted to a trusted /32 address; password authentication is disabled.
- Public and private subnets use separate Network ACLs.
- S3 static content remains private; Block Public Access is enabled.
- S3 server-side encryption and versioning are enabled on the static bucket.
- IAM groups separate Operations, Security, Audit and Intern responsibilities.
- EC2 uses an IAM role instead of stored AWS access keys.
- CloudWatch provides infrastructure metrics, alarms, dashboard and centralized logs.
- Daily backups are automated and a restore path has been tested.

12. Challenge Constraints and Residual Risks

The challenge intentionally restricts the AWS service set. As a result, this design does not use an Application Load Balancer, Auto Scaling Group, Route 53, NAT Gateway, AWS Backup, Systems Manager, CloudFormation or Terraform. A single production t3.micro remains a single point of failure and is not a realistic capacity design for 100,000+ simultaneous users. The tested Golden AMI recovery reduces recovery time but does not provide automatic high availability.

HTTPS port 443 is permitted by network policy, but a publicly trusted TLS termination service/certificate workflow was not established within the recorded implementation. The project should not claim end-to-end HTTPS unless separately configured and verified.

The timed RTO evidence supplied for the project is 3 minutes 9 seconds. This demonstrates the tested recovery run; actual recovery time can vary with AWS capacity, operator execution and the amount of data to restore.

13. Final Project Status

Area	Status
Networking	COMPLETE
Security Group / NACL hardening	COMPLETE
Production EC2 / Apache	COMPLETE
IAM least privilege / MFA	COMPLETE
S3 static content security and integration	COMPLETE
Daily S3 backup	COMPLETE
CloudWatch metrics / alarms / dashboard	COMPLETE
Centralized logging / retention	COMPLETE
Golden AMI DR	COMPLETE
S3 restore test	COMPLETE
RTO <15 min timed test	PASS - 3m 09s
Architecture / RCA / Recovery documentation	COMPLETE

14. Evidence Checklist for Submission

☐ VPC, subnet, route table and Internet Gateway screenshots.

☐ Public and Private NACL rules and subnet associations.

☐ AmazonKart-Web-SG final inbound rules.

☐ Production EC2 details and browser website screenshot.

☐ SSH daemon output proving pubkeyauthentication yes / passwordauthentication no.

☐ IAM groups, users, MFA and EC2 role/policy screenshots.

☐ S3 static bucket Versioning, SSE-S3 and Block Public Access evidence.

☐ Private S3-to-EC2 static image retrieval and browser rendering.

☐ Backup cron entry, crond active output and S3 backup object.

☐ CloudWatch dashboard and CPU/Memory/Disk alarm screenshots.

☐ Four CloudWatch log groups and 30-day retention evidence.

☐ Golden AMI Available screenshot.

☐ DR replacement instance, SSH, HTTP 200 and public browser screenshot.

☐ S3 backup restore commands and post-restore HTTP 200.

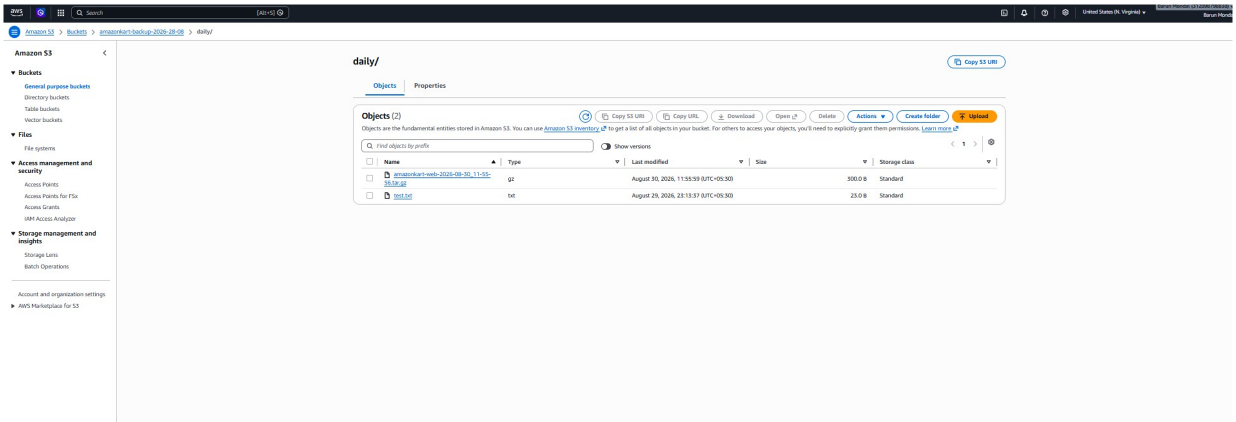
☐ Timed DR test record: 3 minutes 9 seconds.

15. Evidence Appendix

This appendix contains the implementation evidence captured during the build and disaster-recovery validation. Console screenshots are included where they were supplied, and terminal/console outputs are recorded verbatim or as concise evidence extracts.

E1. S3 Daily Backup Object - PASS

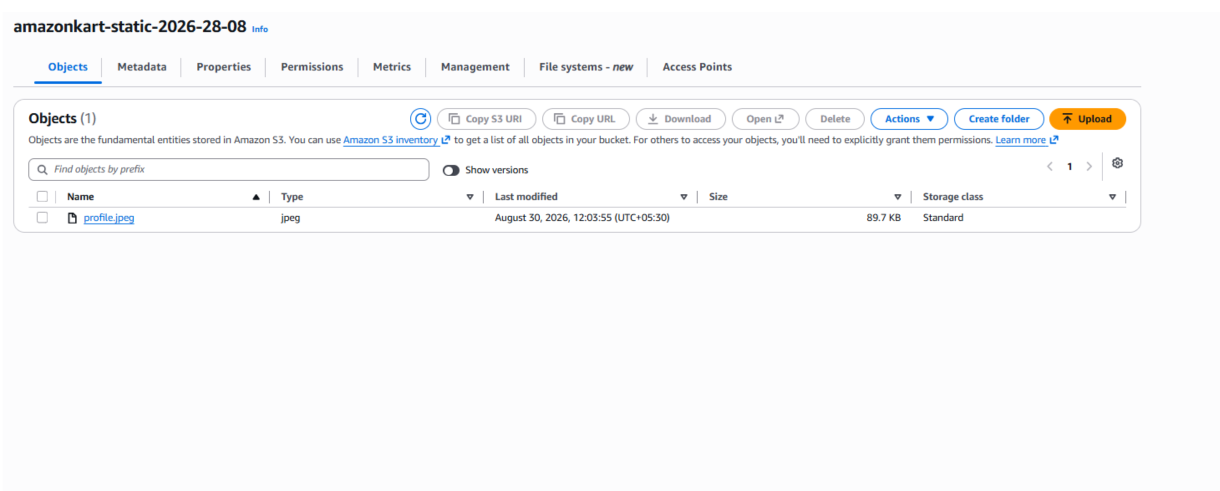
The backup bucket contains the timestamped website archive under the daily/ prefix.




```
Terminal verification:
sudo systemctl is-active crond
active

aws s3 ls s3://amazonkart-backup-2026-28-08/daily/
2026-08-30 11:55:59        300 amazonkart-web-2026-08-30_11-55-56.tar.gz
2026-08-29 23:13:37         23 test.txt

Cron:
0 0 * * * /usr/local/bin/amazonkart-backup.sh
```

E2. S3 Static Object Upload - PASS

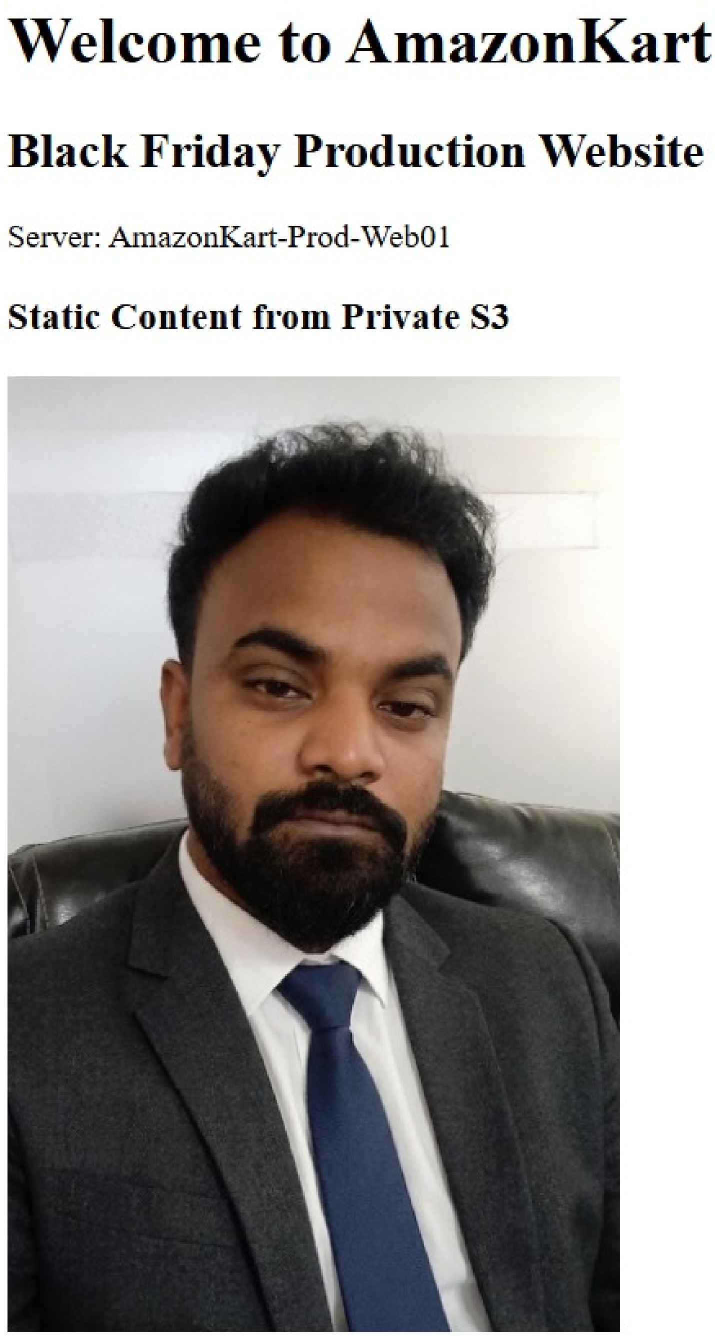
profile.jpeg was uploaded to the private static-content bucket and later retrieved by the EC2 IAM role.



```
EC2 retrieval:
sudo aws s3 cp s3://amazonkart-static-2026-28-08/profile.jpeg /var/www/html/static/profile.jpeg
download: s3://amazonkart-static-2026-28-08/profile.jpeg to ../../var/www/html/static/profile.jpeg

ls -lh /var/www/html/static/profile.jpeg
-rw-r--r--. 1 root root 90K Aug 30 12:03 /var/www/html/static/profile.jpeg
```

E3. Production Website with Private-S3 Static Content - PASS

The production Apache website renders the static image retrieved from the private S3 bucket.




```
curl -I http://localhost/static/profile.jpeg
HTTP/1.1 200 OK
```

```
Server: Apache/2.4.68 (Amazon Linux)
Content-Length: 91856
Content-Type: image/jpeg
```

E4. S3 Static Bucket Security - PASS

Console verification:

Bucket: amazonkart-static-2026-28-08

Region: us-east-1

Bucket Versioning: Enabled

Default encryption: SSE-S3

Block all public access: On

S3 static website hosting: Disabled

E5. Web Security Group Hardening - PASS

```
Security Group: sg-0afcf9ae96a2fd000 / AmazonKart-Web-SG

Final inbound rules:
HTTP   TCP 80   0.0.0.0/0
HTTPS  TCP 443  0.0.0.0/0
SSH    TCP 22   223.228.29.178/32

The previous extra SSH /32 rule was removed. No SSH 0.0.0.0/0 rule remains.
```

E6. Public and Private NACL Segmentation - PASS

```
Public NACL:
acl-0e4b7edca14577380 / AmazonKart-Public-NACL
Associated subnet: AmazonKart-Public-Subnet-1 (10.0.1.0/24)

Public inbound:
100 SSH 22          223.228.29.178/32 ALLOW
110 HTTP 80         0.0.0.0/0         ALLOW
120 HTTPS 443       0.0.0.0/0         ALLOW
130 TCP 1024-65535  0.0.0.0/0         ALLOW
*   All traffic     0.0.0.0/0         DENY

Private NACL:
acl-042bd5c057b67bf13 / AmazonKart-Private-NACL
Associated subnet: subnet-0607efd4f4f728ed4 / AmazonKart-Private-Subnet-1 (10.0.2.0/24)

Private subnet is no longer associated with the Public NACL.
```

E7. CloudWatch Monitoring and Logging - PASS

```
Custom namespace: AmazonKart/Production

Alarms:
AmazonKart-CPU-High-75      CPUUtilization > 75%
AmazonKart-Memory-High-80   mem_used_percent > 80%
AmazonKart-Disk-High-85     disk_used_percent > 85%

Dashboard:
```

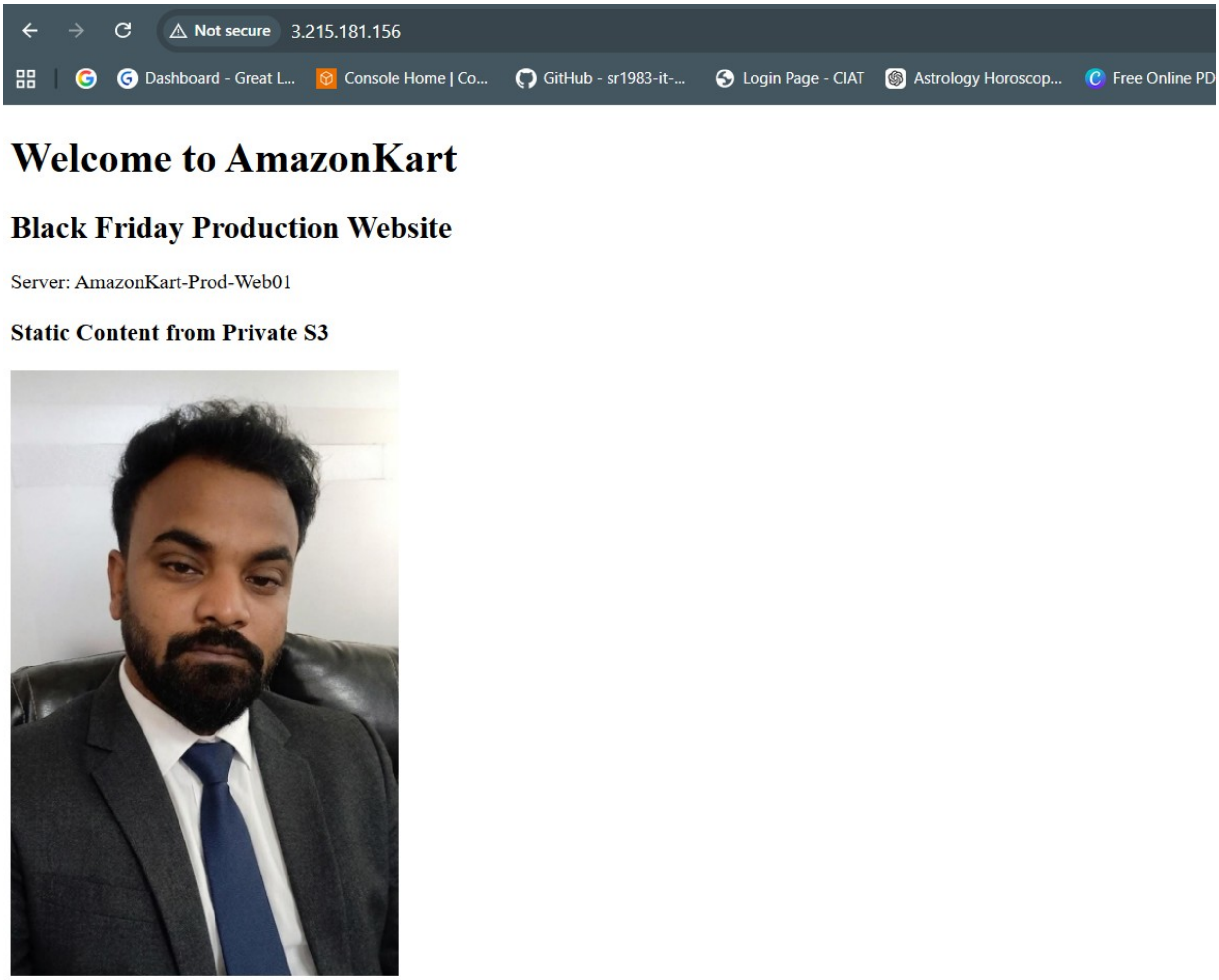
```
AmazonKart-Production-Dashboard

Centralized log groups:
 /amazonkart/production/apache/access
 /amazonkart/production/apache/error
 /amazonkart/production/auth
 /amazonkart/production/system

Retention: 1 month (30 days)
```

E8. Golden AMI for Disaster Recovery - PASS

```
AMI name: AmazonKart-Prod-Golden-AMI
AMI ID: ami-05f02a718c6d21b63
Visibility: Private
Status: Available
Root device type: EBS
Root snapshot: snap-086a6ce085619ab18
Source AMI: ami-0332d564d76dbd8d6
Region: us-east-1
```

E9. DR Replacement Website - Public Browser Test - PASS

A replacement EC2 instance launched from the Golden AMI successfully served the AmazonKart website and static image over its public endpoint.




```
DR instance public endpoint used during test: 3.215.181.156

sudo systemctl is-active httpd
active

curl -I http://localhost
HTTP/1.1 200 OK

curl -I http://localhost/static/profile.jpeg
HTTP/1.1 200 OK
```

E10. S3 Backup Restore on DR Instance - PASS

```
Initial restore test identified missing GetObject permission:
fatal error: An error occurred (403) when calling the HeadObject operation: Forbidden

Corrective IAM action:
Added s3:GetObject for:
arn:aws:s3:::amazonkart-backup-2026-28-08/daily/*

Retest:
aws s3 cp s3://amazonkart-backup-2026-28-08/daily/amazonkart-web-2026-08-30_11-55-56.tar.gz /tmp/
download: ... to ../../tmp/amazonkart-web-2026-08-30_11-55-56.tar.gz

sudo tar -xzf /tmp/amazonkart-web-2026-08-30_11-55-56.tar.gz -C /var/www
```

```
curl -I http://localhost
HTTP/1.1 200 OK
```

E11. Timed Disaster Recovery RTO - PASS

```
Target RTO: < 15 minutes
Recorded elapsed time: 03:09:12
Interpreted project test duration: approximately 3 minutes 9 seconds
Result: PASS

Recovery path:
Golden AMI -> replacement EC2 -> Apache/website available -> public endpoint verified
```

E12. SSH Hardening - PASS

```
sshd effective configuration:
pubkeyauthentication yes
passwordauthentication no

Result:
Key-based SSH enabled.
Password-based SSH disabled.
```

16. Evidence Result Summary

Evidence ID	Control / Test	Result
E1	Daily S3 backup	PASS
E2	Private S3 static object retrieval	PASS
E3	Website static content rendering	PASS
E4	S3 versioning/encryption/public-access controls	PASS
E5	Security Group hardening	PASS
E6	Public/Private NACL separation	PASS
E7	CloudWatch monitoring/logging	PASS
E8	Golden AMI	PASS
E9	DR public website recovery	PASS
E10	S3 backup restore	PASS
E11	RTO <15 minutes	PASS - ~3m 09s
E12	SSH key-only hardening	PASS